Nandigama Vaishnavi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.54
NARAYANA JUNIOR COLLEGE,VELIMELA, SANGAREDDY	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2024	98.10%
KRISHNAVENI TALENT SCHOOL,PARIGI, VIKARABAD	Class X (BOARD OF SECONDARY EDUCATION TELANGANA)	2022	10.00%

POSITIONS OF RESPONSIBILITY

PR & Outreach Head | Prayas Club, IIT Hyderabad

- Led outreach initiatives and coordinated promotional campaigns for club events.
- Collaborated with schools and external organizations to increase participation.

Outreach Coordinator | ELAN & nVision, IIT Hyderabad

- Coordinated outreach activities and promoted flagship fest events.
- Worked with student ambassadors and organizing teams.

Hostel Representative | Sarojini Naidu Hostel, IIT Hyderabad

- Coordinated hostel activities and student communication.
- Served as a liaison between residents and the hostel administration.

SKILLS

Languages & Core: C, C++, Python, JavaScript (ES6+), Data Structures & Algorithms

Frontend: HTML, CSS, React.js, React Router, Tailwind CSS, Axios, Context API

Backend: Node.js, Express.js, RESTful API Design

Database: MongoDB, Mongoose

Auth & Security: JWT, bcrypt, Role-Based Access Control

Tools & Platforms: Git, GitHub, Postman, Cloudinary, MongoDB Atlas, Render, Vercel

Soft Skills: Communication, Leadership, Teamwork, Problem Solving

RELEVANT COURSES

Data Structures & Algorithms,Linear Algebra,Probability & Statistics,Calculus,Differential Equations

PROJECTS

Coupon -- Verified Coupon Marketplace

Live Demo | Frontend Repo | Backend Repo

React, Node.js, Express, MongoDB, JWT, Cloudinary

- Built and deployed a full-stack marketplace enabling users to buy/sell verified discount coupons, with JWT-based authentication and role-based access control (buyer, seller, admin)
- Designed an atomic purchase system using MongoDB's findOneAndUpdate to prevent race conditions, ensuring a coupon cannot be sold to multiple buyers simultaneously
- Implemented a secure code-reveal mechanism using schema-level field exclusion (select: false), hiding sensitive coupon codes until payment is confirmed
- Built an admin verification workflow to approve/reject seller-submitted coupons before public listing, with image uploads handled via Cloudinary
- Developed RESTful APIs with layered architecture (routes/controllers/services/models), input validation, and centralized error handling
- Deployed backend on Render and frontend on Vercel, with MongoDB Atlas as the cloud database

Whitworth Quick-Return Mechanism Simulator | Python, NumPy, Matplotlib, Tkinter

- Developed an interactive **Whitworth quick-return mechanism simulator** with real-time animation, playback controls, configurable parameters, and mechanism visualization.
- Implemented **kinematic analysis of position, velocity, and acceleration**, including loop-closure residual checks and three-position orientation analysis.
- Performed **dynamic torque and crank-pin force analysis using the Virtual Power Method**, incorporating mass properties, moments of inertia, and material parameters.

Four-Bar & Slider-Crank Mechanism Simulation | Python, NumPy, Matplotlib

- Developed a simulation program for **four-bar and slider-crank mechanism inversions**, including Grashof/non-Grashof classification and mechanism animation.
- Implemented **position, velocity, and acceleration analysis** for mechanism links and arbitrary points on the coupler.
- Simulated different configurations and visualized **coupler-point kinematics and slider motion** for prescribed angular inputs.

EXTRACURRICULAR

Active participation in social outreach initiatives and community service programs.

Volunteered in activities focused on education, mentorship, and social awareness.

Participated in campus events and collaborative student activities.